

The Rooted Knauf Clock Does Not Descend to Cycles

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Abstract

The Knauf binary recursion has the source-owned partition limit $\sum_{n \geq 1} \varphi(n)n^{-s} = \zeta(s-1)/\zeta(s)$ for $\text{Re } s > 2$. We ask the narrower typed question of whether its exact rooted label $h(w) = \mathbf{1}^\top M_w e_1$ also supplies a cyclic primitive ledger, additive repetition clock, scalar Liouville phase, and a Fredholm determinant whose trace powers enumerate those same primitive returns. It does not. The trailing-zero stable quotient admits no canonical right append-one action; $h(01) \neq h(10)$; and $h(11) \neq h(1)^2$. Moreover, for the Liouville function $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n)$ counts prime factors with multiplicity, $\lambda(h(w))$ fails both cyclic and power descent. A diagonal operator on stable states does realize the partition function as a trace and owns an ordinary marked determinant, but its marker counts inventory powers rather than binary returns. These exact witnesses yield a source-specific Route-A rejection while leaving trace clocks, enlarged states, non-scalar cocycles, and established Farey/Gauss transfer models outside the theorem. The selection and witness package are retrospective and carry no prospective or priority claim.

Claim boundary. This paper closes only the frozen SD-C06 test for the rooted label $h(w) = \mathbf{1}^\top M_w e_1$, its trailing-zero colimit, ordinary binary necklaces and word powers, the literal scalar Liouville observable, and the diagonal state inventory. It makes no universal claim about enlarged Knauf, Farey, Gauss, Selberg, adelic, trace-clock, or cocycle models. The selection is retrospective, the strict disposition is ROUTE_A_REJECTED, and Route B is locked.

1 Introduction

An arithmetic partition function can resemble a dynamical zeta expression without carrying the primitive objects that a dynamical determinant counts. The distinction is especially sharp for the rooted Knauf binary recursion: its limiting partition sum is an exact quotient of zeta functions, while its finite source states are rooted words embedded by trailing zeroes. The quotient identity and its phase-transition setting belong to the original number-theoretic spin-chain literature (Knauf, 1993, 1998, 1999).

The issue studied here is not whether that arithmetic identity is valid. It is whether the very same finite object and label also determine the data needed for a scalar primitive-orbit determinant: a state on which the binary step acts autonomously, a clock that is independent of the root of a cyclic word and additive under repetition, a scalar phase with the corresponding power law, and an operator whose trace powers count those same returns. A partition trace supplies none of these identifications by notation alone.

We therefore evaluate the smallest object-preserving bridge. We retain the source matrices, the rooted label $h(w) = \mathbf{1}^\top M_w e_1$, the stable-state quotient under trailing zeroes, ordinary cyclic rotation, ordinary word powers, and the literal Liouville observable $\lambda(h(w))$, where $\lambda(n) = (-1)^{\Omega(n)}$ and $\Omega(n)$ counts prime factors with multiplicity. Exact words of length at most three then settle the necessary descent questions; Figure 1 previews the object-preserving bridge and its first failure. A separate trace-class argument identifies the diagonal determinant that the stable-state inventory actually owns. Neither argument uses target zeros, fitted parameters, or a prime-indexed external component.

Related Farey constructions make the typing issue concrete. A translation-invariant Farey chain can share free energy with a related non-translation-invariant number-theoretic chain, and generalized Farey models connect matrix recursions to intermittent transfer operators (Kleban and Ozluk, 1999; Fiala et al., 2003; Fiala and Kleban, 2005). Trace-energy chains exploit cyclic invariance, and prescribed-trace matrix products have their own arithmetic counting theory (Prellberg et al., 2006; Technau, 2023). These are genuine models, but equality of free energy or use of the same generators does not identify their finite objects, clocks, or return operators with the rooted first-column-sum object tested here.

The same firewall applies to determinant language. Orbit zeta and transfer-operator determinants are genuine dynamical constructions when their phase spaces, weights, and operator domains are specified (Ruelle, 1976; Mayer, 1990, 1991); their existence does not identify an operator for the rooted label. Conversely, the diagonal operator constructed below is a valid Fredholm object, but its free marker enumerates powers of inventory eigenvalues rather than binary returns.

The contribution is therefore a closure theorem, not a new zeta mechanism. It has three parts. First, the canonical append-one step does not descend to the trailing-zero colimit. Second, the rooted clock and literal scalar phase fail cyclic and repetition descent by explicit witnesses. Third, the exact state-inventory determinant is derived and assigned to its legal owner. The source recursion and quotient, thermodynamic trace-chain repairs, generalized transfer operators, and the general partition-trace principle remain prior art. The selection that led to this test is retrospective and receives no novelty or priority credit.

Section 2 records this source ownership and the claim boundary. Section 3 fixes the types and matrix convention. Section 4 proves the finite non-descent theorems, and Section 5 derives the determinant owned by state inventory. Section 6 maps those results to the strict Route coordinates without adding experiment claims. Complete exact products, domain restrictions, and literature firewalls appear in the appendices.

2 Prior ownership, retrospective selection, and scope

Source arithmetic. The source literature owns the number-theoretic spin chain, its phase structure, the stable multiplicity $\varphi(n)$, and the quotient $\zeta(s-1)/\zeta(s)$. The partition-function and phase analysis is already present in Knauf (1993); the binary arithmetic chain and its Riemann-zero program, together with the required correction, are source work of Knauf (1998, 1999). We inherit those facts and use the source matrix convention. We do not treat the quotient, the totient ledger, or the phase transition as a Paper-41 result.

Thermodynamic equivalence is not object identity. The Farey fraction chain of Kleban and Ozluk (1999) is translation invariant and has the same free energy as a related non-translation-invariant number-theoretic chain. Several Farey-fraction models likewise share free energy and connect to transfer-operator thermodynamics (Fiala et al., 2003). From these source facts we draw only a typing inference: equality of free energy does not imply equality of finite states, rooting convention, clock, primitive ledger, marker, or operator. This sentence is not attributed to those papers as a quoted theorem; it is the contract discipline used in the present audit.

Changed trace and transfer models. Parameterized generalized chains are connected to an intermittent transfer operator and the Lewis equation (Fiala and Kleban, 2005). A trace-energy Farey chain obtains translation invariance from cyclicity of the matrix trace, while its matrices and energies remain distinguished from generalized Knauf quantities even when thermodynamic limits agree (Prellberg et al., 2006). The adelic Markov operator studied by Knauf (2013) is a further, separate operator and phase space. None of these changed models acts by declaration on the frozen rooted-state quotient.

Recent trace counting. [Technau \(2023\)](#) counts products of Farey matrices with prescribed trace. That result is relevant positive context for trace-based arithmetic, but trace is conjugacy-compatible whereas $h(w) = \mathbf{1}^\top M_w e_1$ reads a distinguished column. It therefore does not settle the rooted non-descent theorem below.

Table 1: Ownership firewall. The final column records whether source ownership transfers to the exact SD-C06 rooted contract.

Source family	Finite object	Clock or energy	Operator owner	Transfer to SD-C06
Exact Knauf rooted chain	Rooted binary word and trailing-zero stable class	$h(w)$ and $\log h(w)$	Finite inventory; no declared autonomous return operator	Source-owned object; descent must be proved
Farey trace chain	Cyclic or translation-invariant matrix word	$\text{Tr}(M_w)$ -based energy	Trace/Farey model operator	Changed clock and object
Generalized chain	Parameterized recursive state	Generalized energy/expectation data	Intermittent transfer model	Changed object and operator
Adelic Markov model	Adelic state space	Markov spectral data	Adelic transition operator	Separate phase space
Gauss/Selberg family	Continued-fraction or modular returns	Derivative/geodesic weights	Dynamical transfer operator	Comparator only
Paper-41 diagonal inventory	Stable states indexed with multiplicity $\varphi(n)$	n^{-s} eigenvalue weight	Q_s on $\ell^2(S_K)$	Owens only its inventory determinant

The historical candidate was selected by a Boolean rule applied after all six Session-4 cards, their outcomes, and the present witnesses were known. The unique rule result is SD-C06. The rule required the exact card statuses AO_ANALYTIC_ARITHMETIC_ORIGIN, A1_FAIL, A2_FAIL, and A3_PARTIAL_ANALYTIC_STRUCTURE, together with the card’s request for a primitive-ledger no-go and an endogenous-sign test. Paper 39 supplies existence provenance but performs no ranking or successor authorization; Paper 40 supplies neither selection nor novelty. Uniqueness of this retrospective rule is not prospective evidence and does not establish discovery priority.

Our quantifiers stop at declared source-preserving attempts. Enlarging the state to the full matrix, replacing h by trace or an eigenvalue, adding a history-dependent cocycle, or importing a Farey/Gauss transfer operator is a legitimate new model but not an in-place repair. The bounded literature audit did not locate the exact four-witness conjunction in a primary source; that is a search result rather than a universal novelty theorem. All source modules in [Table 1](#) retain their own credit.

3 Frozen rooted object and typed quotients

We fix the source convention of the Knauf chain ([Knauf, 1998, 1999](#)). Let $W = \{0, 1\}^*$, with empty word ε , and set

$$L = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad M_w = M_{w_1} \cdots M_{w_k},$$

where $M_0 = L$ and $M_1 = R$. The rooted label is

$$h(w) = \mathbf{1}^\top M_w e_1, \quad \mathbf{1} = (1, 1)^\top, \quad e_1 = (1, 0)^\top.$$

The citation fixes source ownership and convention; the calculation below is an explicit derivation rather than an appeal to authority.

Lemma 3.1 (Recurrence fidelity). *For a binary word u , let $\bar{u}$ denote its bitwise complement. Then*

$$h(u0) = h(u), \quad h(u1) = h(u) + h(\bar{u}).$$

Proof. Write $M_u = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then $h(u) = a + c$. Right multiplication gives

$$M_u L = \begin{pmatrix} a & a+b \\ c & c+d \end{pmatrix}, \quad M_u R = \begin{pmatrix} a+b & b \\ c+d & d \end{pmatrix}.$$

Thus $h(u0) = a + c$ and $h(u1) = a + b + c + d$. With $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ one has $JLJ = R$, $JRJ = L$, and hence $M_{\bar{u}} = JM_uJ$. It follows that $h(\bar{u}) = b + d$, proving the recurrence. $\square$

This identity matters because h is rooted. It reads the first column of M_w , whereas $\text{Tr}(M_w)$ is invariant under cyclic permutation of factors. The latter property underlies trace-energy Farey models (Prellberg et al., 2006); it is a positive control, not an identity for the rooted label.

The relevant mathematical types are listed in Table 2. The table prevents three common category errors: equating the trailing-zero colimit with a shift space, treating a necklace as a rooted word with forgotten notation, and interpreting a free Fredholm marker as physical return time.

Table 2: Object, equivalence, marker, and ownership types.

Object	Equivalence or indexing	Marker	Legal data/operator	Prohibited identification
KnaufRootedWord	Equality of rooted words	Depth k	M_w , first column, $h(w)$	Cyclic primitive class
KnaufStableState	Colimit relation generated by $w \sim w0$	Dirichlet variable s	Stable h and full multiplicity ledger	Autonomous right append action
BinaryNecklace	Cyclic rotation of a nonempty word	Repetition r	Period and powers after a clock descends	Rooted h value
FareyTraceWord	Matrix/cyclic word	Trace or eigenvalue clock	Cyclic trace and matrix-power data $\lambda(h(w))$	Frozen rooted partition trace
LiouvilleState Observable	Arithmetic value after evaluating h	Scalar sign		Endogenous symbolic cocycle
StateInventory Diagonal	Stable-state basis with multiplicity	Free marker u	Q_s and its marked determinant	Binary return operator
DynamicalTransfer Operator	Separately declared phase space	Return/iterate marker	Trace powers only when constructed	Source ownership by notation

In particular, depth k , Dirichlet variable s , determinant marker u , and temporal repetition r have different owners. No result below identifies them.

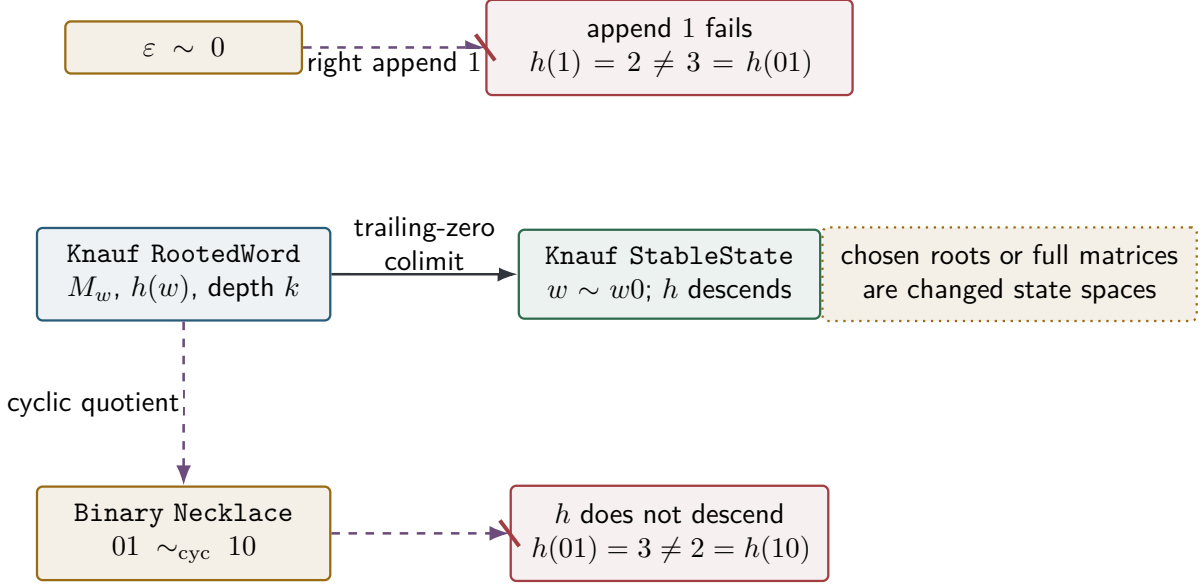


Figure 1: Typed source quotients. The trailing-zero relation preserves h , but the would-be append-one images of $\varepsilon \sim 0$ have labels 2 and 3, so append-one does not descend. Cyclic descent fails independently on $01 \sim_{\text{cyc}} 10$. Full-matrix states or chosen representatives change the contract.

4 Exact non-descent theorem

All witnesses in this section use the source-owned matrix convention (Knauf, 1998, 1999). The products are reproduced in Appendix A; no literature citation substitutes for their exact evaluation.

Lemma 4.1 (Witness ledger). *The frozen label satisfies*

$$h(\varepsilon) = 1, \quad h(0) = 1, \quad h(1) = 2, \quad h(01) = 3, \quad h(10) = 2, \quad h(11) = 3,$$

and

$$h(001) = 4, \quad h(010) = 3.$$

Proof. These values follow by direct multiplication of L and R . For example, $LR = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and $RL = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, so their first-column sums are 3 and 2. The remaining products are displayed in Table 4. $\square$

Theorem 4.2 (Rooted-state and clock non-descent). *There is no map on the trailing-zero quotient satisfying $A_1([w]) = [w1]$ for every rooted word. There is no function on binary necklaces whose value on $[w]_{\text{cyc}}$ is $h(w)$ for every nonempty word. Independently, the identity $h(w^r) = h(w)^r$ does not hold for every word and repetition.*

Proof. Let $\sim_0$ be the equivalence relation generated by $w \sim_0 w0$. The recurrence gives $h(w0) = h(w)$, so h is constant on every stable class. Yet $[\varepsilon] = [0]$ while their proposed append-one images would be $[1]$ and $[01]$. Because $h(1) = 2 \neq 3 = h(01)$, these images lie in distinct stable classes. Thus the proposed A_1 is not well defined.

For cyclic descent, 01 and 10 are rotations, but $h(01) = 3 \neq 2 = h(10)$. A representative-independent necklace function cannot take both values. For temporal powers, the single-letter word gives $h(11) = 3 \neq 4 = h(1)^2$. These failures are logically independent of the first quotient failure. Since all values are positive, the last two equalities are also necessary for $T = \log h$ to be a cyclic and temporally additive orbit clock. $\square$

Remark 4.3 (Quantifier boundary). The first assertion excludes the canonical right append-one rule on the source colimit, not every self-map of the underlying set. The second and third concern the exact h clock, cyclic rotation, and ordinary word powers. They do not exclude a chosen-root convention, a full-matrix state, a trace clock, or an eigenvalue clock; each changes at least one frozen coordinate.

Theorem 4.4 (Literal scalar-phase non-descent). *The observable $\ell(w) = \lambda(h(w))$ neither descends to binary necklaces nor satisfies $\ell(w^r) = \ell(w)^r$ for all words and repetitions. No one-letter multiplicative character equals ℓ on every rooted word.*

Proof. The cyclic rotations 001 and 010 have labels 4 and 3. Hence $\ell(001) = \lambda(4) = +1$ whereas $\ell(010) = \lambda(3) = -1$. For repetition, $\ell(1) = \lambda(2) = -1$ while $\ell(11) = \lambda(3) = -1 \neq (+1) = \ell(1)^2$.

A one-letter multiplicative character has the form $\chi(w) = \alpha^{\#0(w)}\beta^{\#1(w)}$. Agreement with ℓ on 0 and 1 forces $\alpha = 1$ and $\beta = -1$. Agreement on 11 would then require $-1 = \ell(11) = \beta^2 = +1$, a contradiction. $\square$

The theorem is about the literal scalar observable evaluated after h . It does not quantify over arbitrary history-dependent or matrix-valued cocycles on enlarged states. That distinction is essential: calling $\lambda(h(w))$ endogenous does not make it a symbolic orbit character.

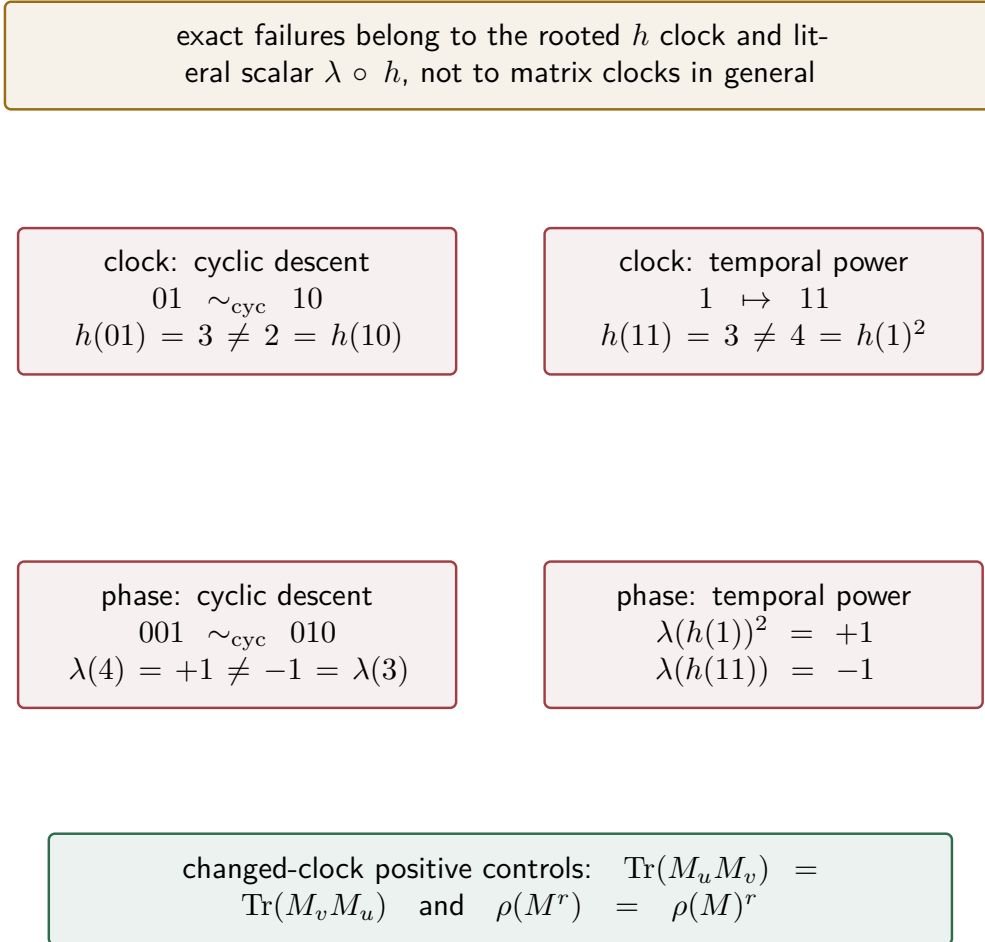


Figure 2: Exact clock and phase witnesses. Cyclic rotation and ordinary powers fail for the frozen rooted label and literal Liouville observable. The positive-control strip records that matrix traces and eigenvalue clocks can pass analogous tests only after the clock type has changed.

The positive controls show that the argument is projection-specific rather than a blanket no-go. Cyclic trace models are established in the Farey literature, and prescribed-trace products form

a valid arithmetic family (Prellberg et al., 2006; Technau, 2023). Algebraically, $\text{Tr}(M_u M_v) = \text{Tr}(M_v M_u)$, and the Perron eigenvalue of a fixed matrix obeys the matrix-power law. Those controls change the rooted clock and therefore do not repair Theorems 4.2 and 4.4 in place.

5 The determinant owned by state inventory

Let S_K be the trailing-zero colimit. We inherit the source multiplicity theorem

$$\#\{x \in S_K : h(x) = n\} = \varphi(n)$$

and the associated partition identity from the number-theoretic spin-chain literature (Knauf, 1993, 1998, 1999). The following theorem does not create primitive dynamics; it identifies the operator and determinant already forced by that inventory.

Theorem 5.1 (State-inventory determinant). *For $\text{Re } s > 2$, define the diagonal operator on $\ell^2(S_K)$ by*

$$Q_s e_x = h(x)^{-s} e_x.$$

Then Q_s is trace class and

$$\text{Tr } Q_s = \sum_{n \geq 1} \varphi(n) n^{-s} = \frac{\zeta(s-1)}{\zeta(s)}.$$

Its Fredholm determinant is entire in the free variable u and equals

$$\Delta_K(s, u) = \det(I - uQ_s) = \prod_{n \geq 1} (1 - u n^{-s})^{\varphi(n)}.$$

For $|u| < 1$ its local trace-log is

$$-\log \Delta_K(s, u) = \sum_{r \geq 1} \frac{u^r}{r} \frac{\zeta(rs-1)}{\zeta(rs)}.$$

Finally, $\Delta_K(s, 1) = 0$.

Proof. Put $\sigma = \text{Re } s > 2$. Because Q_s is diagonal, its trace norm is

$$\|Q_s\|_1 = \sum_{x \in S_K} |h(x)^{-s}| = \sum_{n \geq 1} \varphi(n) n^{-\sigma} < \infty.$$

Thus Q_s is trace class. Absolute convergence permits regrouping by the exact multiplicity of each label, yielding the displayed zeta quotient.

Trace-class Fredholm theory gives an entire function of u and the product over eigenvalues. Regrouping those eigenvalues by n gives the second product. Moreover $\|Q_s\| = 1$, since the $n = 1$ eigenvalue is one and every other eigenvalue has modulus below one. Hence, for $|u| < 1$, the operator series for $-\log(I - uQ_s)$ converges in the domain needed to take traces. Therefore

$$-\log \det(I - uQ_s) = \sum_{r \geq 1} \frac{u^r}{r} \text{Tr}(Q_s^r).$$

The same multiplicity ledger gives

$$\text{Tr}(Q_s^r) = \sum_{n \geq 1} \varphi(n) n^{-rs} = \frac{\zeta(rs-1)}{\zeta(rs)},$$

because $\text{Re}(rs) > 2$ for every positive integer r . Finally, $\varphi(1) = 1$, so the determinant product contains the factor $1 - u$ and vanishes at $u = 1$. $\square$

The theorem separates three statements that can otherwise be conflated. The coefficient of u in the trace-log is the source partition function. The higher coefficients count powers of fixed diagonal eigenvalues. The free marker u is neither binary depth nor temporal return time, and Q_s depends on the Dirichlet parameter rather than defining a fixed self-adjoint Hilbert–Polya operator.

This diagonal inventory determinant is not the orbit determinant appearing in expanding-map or Gauss/modular transfer-operator formalisms (Ruelle, 1976; Mayer, 1990, 1991). Its marker enumerates powers of fixed inventory eigenvalues, not returns of binary primitive classes. Those references provide changed-model context; they do not prove the formula for Q_s , which was derived above.

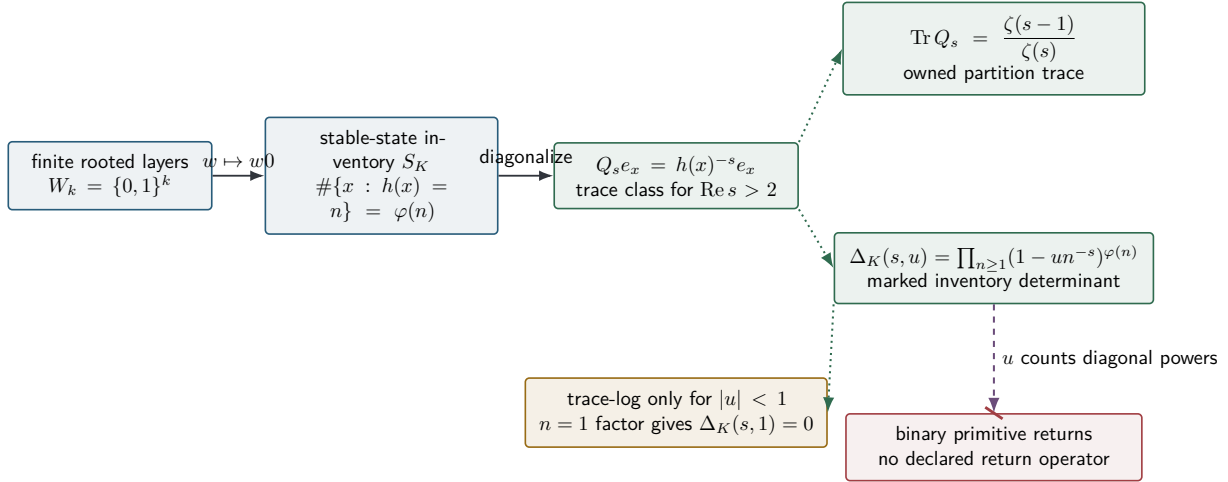


Figure 3: The stable-state inventory owns both the partition trace and its marked diagonal Fredholm determinant on $\text{Re } s > 2$, with the displayed trace-log restricted to $|u| < 1$. The marker counts diagonal powers, not binary primitive returns; the barred arrow prevents that ownership from being transferred to the missing dynamical operator.

Thus Theorem 5.1 is an ownership control. It proves that the obstruction is not an absence of all Fredholm determinants. It also proves why this particular determinant cannot earn A2 primitive-orbit credit: the source binary refinement has no declared return operator on S_K , and the rooted clock already fails the cyclic and power laws required for such a ledger.

6 Strict Route audit and reproducibility boundary

The strict Route tuple is a typed summary of the preceding proofs, not a numerical score. Each coordinate has an independent owner and failure mode, as shown in Table 3.

Consequently the strict tuple is

$$(\text{AO_ANALYTIC_ARITHMETIC_ORIGIN}, \text{A1_FAIL}, \text{A2_FAIL}, \\ \text{A3_PARTIAL_ANALYTIC_STRUCTURE}, \text{A4_FAIL})$$

with overall verdict `ROUTE_A_REJECTED` and Route B disallowed. The positive source identity does not override the object failures in A1 and A2, and the parameter-dependent diagonal operator does not upgrade A4.

The independent DA verified the sealed research package, typed source resolver, matrix convention, exact witnesses, theorem quantifiers, operator domains, retrospective selection, literature boundary, and Route coordinates. That review is governance evidence for the frozen research bytes; it is not a numerical experiment and does not enlarge the theorem.

Table 3: Coordinate-by-coordinate Route evidence.

Rung	Retained evidence	Decisive boundary	Status
A0	Source multiplicities give the exact arithmetic partition quotient on $\text{Re } s > 2$	The quotient alone supplies neither primitive rational-prime objects nor an endogenous sign cocycle	A0_ANALYTIC ARITHMETIC_ORIGIN
A1	Matrix convention and all finite products are exact	Append-one, cyclic clock, temporal power, and scalar-phase descent fail	A1_FAIL
A2	Q_s owns a valid marked Fredholm determinant	Its states and marker are inventory powers, not primitive binary returns	A2_FAIL
A3	The source quotient has a displayed meromorphic continuation	No same-object completed divisor, functional equation, or signed critical-half-plane theorem is derived	A3_PARTIAL ANALYTIC STRUCTURE
A4	A parameter-dependent diagonal trace operator exists	No fixed self-adjoint operator, same-clock trace identity, or target multiplicity theorem exists	A4_FAIL

Canonical authority integration: FINAL / POST-OUTPUT CLEAN. The matrix-primary and recurrence-independent evaluators passed 24/24 and 25/25 checks. Their scientific projections are byte-identical; the canonical projection has SHA-256 `f9cbcd9a757896b976ad81a66f235d670029f727b6fda9b4e851846bac50bec`. The portable resolver matched 22/22 typed source IDs, the theorem-failure count is zero, and the retrospective selector returned only SD-C06.

The main and independent strict Route validators passed 51/51 and 24/24 checks, respectively, with tuple (A0_ANALYTIC_ARITHMETIC_ORIGIN, A1_FAIL, A2_FAIL, A3_PARTIAL_ANALYTIC_STRUCTURE, A4_FAIL), overall ROUTE_A_REJECTED, and Route B false. The adversarial suite rejected all 1168 mutations (packet 288, selection 103, Route 473, static 89, and read-only audit 215), with zero survivors and 1728 dual-evaluator rejection decisions. A/B/cold artifacts and paired audit stdout are byte-identical. The first canonical materialization changed 42 managed paths; the exact idempotence rerun changed zero.

The integrity audit passed 51/51 checks; its SHA-256 is `cf16ba60ca4a45e3e31c5e48590c7cafd69d45694ab343cecf0076476f936d5e`. The exact namespace contains 39 result paths, two evaluation paths, and one experiment report. Its 79-entry result ledger has SHA-256 `25f7aa42c6d55d7608509dbe1b66586bda6db98120ba450f5b59a5c6f3d19f99`. The integration chronology remains RETROSPECTIVE_CORRECTIVE_RESEAL_AFTER_FAILED_OUTPUTS_AND_AUDIT_FINDINGS; these checks reproduce the frozen claims and confer no prospective or novelty credit. In particular, `results_unseen`, `blind`, and `fully_prospective` are all false. This is still the Stage-A artifact: `source_commit`, `code_commit`, and `source_lock.code_commit` remain PENDING_FIRST_ARTIFACT_COMMIT, and `PAPER_MANIFEST.sha256` is absent.

The reproducibility boundary is therefore simple. Theorems 4.2–5.1 depend only on the frozen source convention, exact integer products, the inherited multiplicity theorem, and trace-class algebra. A later implementation can independently replay those dependencies, but it cannot strengthen the claim, change the Route tuple, or transfer ownership from another model without a new source lock.

7 Limitations and conclusion

The Knauf partition quotient remains a rigorous arithmetic trace. What the frozen source does not supply is a canonical cyclic primitive ledger with the same rooted clock, ordinary repetition, literal Liouville phase, and binary return operator. The exact witnesses establish that bounded

statement, while the diagonal construction identifies the determinant that state inventory actually owns.

The conclusion is intentionally local. Matrix traces, Perron eigenvalues, full-matrix states, non-scalar cocycles, and established transfer operators may restore missing structures by changing the model. Trace/Farey repairs and generalized transfer models are legitimate new source locks (Kleban and Ozluk, 1999; Fiala and Kleban, 2005; Prellberg et al., 2006). The adelic Markov operator is a separate alternative (Knauf, 2013), and Gauss/modular transfer operators have their own phase spaces and determinant ownership (Mayer, 1990, 1991). None is an unresolved exception to the theorem proved here.

A reopened candidate must therefore declare, before analytic target comparison, its phase space, equivalence relation, clock, repetition marker, scalar or non-scalar phase, function space, and operator owner. It must then prove the bridge between those coordinates rather than import it from equal free energy or shared matrices. If the rooted h label and its literal types are retained, the present witnesses remain terminal. If a coordinate is changed, the resulting construction should be evaluated as a new model with its own provenance and Route record.

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A Exact algebra and complete proof map

This appendix makes every finite calculation used above explicit. The matrix convention and multiplicity input are source-owned (Knauf, 1998, 1999, 1993); all deductions from them are reproduced here.

A.1 Recurrence and exact products

For $M_u = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, right multiplication by L preserves the first column, while right multiplication by R replaces it by the sum of the two columns. The conjugacy $M_{\bar{u}} = JM_uJ$ identifies the second-column sum with $h(\bar{u})$. This proves $h(u0) = h(u)$ and $h(u1) = h(u) + h(\bar{u})$ without transposition or reversed word order.

The exact products needed by all witnesses are

Table 4: Complete finite witness ledger.

w	M_w	$h(w)$
ε	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	1
0	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	1
1	$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$	2
01	$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$	3
10	$\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$	2
11	$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$	3
001	$\begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$	4
010	$\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$	3

A.2 Stable quotient and rooted action

Let $\sim_0$ be generated by $w \sim_0 w0$. Repeated use of the recurrence shows that h is constant on each equivalence class. Since $\varepsilon \sim_0 0$, a descended right append-one action would imply $[1] = [01]$. The table gives $h(1) = 2$ and $h(01) = 3$, contradicting class invariance. The contradiction uses the same invariant that certifies the source quotient; it does not assume a topology or an undeclared shift map.

A.3 Clock and phase descent

The words 01 and 10 differ by a cyclic rotation, but their rooted labels are 3 and 2. Hence no function on necklaces can equal h on every representative. The power witness is independent: $11 = 1^2$ as a word, yet $h(11) = 3 \neq 4 = h(1)^2$. Because $h > 0$, these identities are necessary for $\log h$ to be representative-independent and additive under powers.

For the literal phase, 001 and 010 are rotations and have labels 4 and 3. Their Liouville values are +1 and -1. Also $\lambda(h(11)) = \lambda(3) = -1$, whereas $\lambda(h(1))^2 = \lambda(2)^2 = +1$. Finally, a one-letter character agreeing on 0 and 1 must have generator values 1 and -1, so it takes the value +1 on 11, contradicting the exact value -1.

A.4 Inventory determinant and repair trilemma

The source multiplicity theorem gives one eigenvalue n^{-s} for each of the $\varphi(n)$ stable states with label n . Absolute summability on $\operatorname{Re} s > 2$ proves trace class, justifies grouping the trace, and yields the product in Theorem 5.1. For $|u| < 1$, the norm-convergent logarithm gives the trace powers. This chain uses no cyclic-word identification. In particular, the $n = 1$ inventory state forces the factor $1 - u$.

The declared source-preserving repair list is now exhausted. Quotienting rooted words by rotation contradicts the clock witness. Keeping the literal scalar phase contradicts the phase

witnesses. Keeping the stable quotient with the full right binary action contradicts append-one non-descent. Using Q_s changes the marker to inventory powers. Trace/eigenvalue clocks and enlarged matrix states change the source object or clock. Thus at least one frozen coordinate must be surrendered; the statement does not range over undeclared extensions.

The positive controls confirm that the proof does not prohibit cyclic or temporal matrix data in general. Direct calculation gives $\text{Tr}(LR) = \text{Tr}(RL) = 3$, and the spectral radius of a fixed positive matrix obeys $\rho(M^r) = \rho(M)^r$. Both controls pass only after replacing the rooted first-column-sum clock.

B Operator, literature, Route, and lock boundaries

B.1 Analytic and operator domains

The determinant statement has distinct domains that are not interchangeable.

Table 5: Analytic domain ledger for the source quotient and inventory determinant.

Statement	Domain	What is not implied
$\sum_n \varphi(n)n^{-s} = \zeta(s-1)/\zeta(s)$	$\text{Re } s > 2$ by absolute convergence	Finite-depth convergence outside the domain
Q_s trace class and $\det(I - uQ_s)$ entire in u	Fixed s with $\text{Re } s > 2$	A fixed self-adjoint operator independent of s
Local trace-log series	$\text{Re } s > 2$ and $ u < 1$	A global logarithm across determinant zeros
Meromorphic continuation of the scalar quotient	As a quotient of zeta functions	Continuation of missing primitive dynamics or signed convergence

At $u = 1$ the determinant vanishes because of the $n = 1$ state. This is not a singularity to be removed in order to recover the partition quotient; it is part of the determinant owned by the inventory. The source partition trace is its first trace-log coefficient, not the determinant evaluated at a special marker.

B.2 Literature collision matrix

The literature audit was bounded by the frozen search protocol. The phrase “the audit did not locate” records that search; it is not a claim that no prior work exists.

Table 6: Source ownership and collision boundary.

Prior family	Source-owned result	Paper-41 boundary
Knauf source chain (Knauf, 1993, 1998, 1999)	Partition quotient, phase setting, recursion, and arithmetic program	Only the exact rooted non-descent conjunction is tested here
Farey free-energy equivalence (Kleban and Ozluk, 1999; Fiala et al., 2003)	Equal free energies for related but distinct formulations	No transfer of finite object, clock, ledger, or operator
Generalized transfer model (Fiala and Kleban, 2005)	Parameterized chain, intermittent operator, and Lewis-equation connection	Changed state and operator
Trace-energy chain (Prellberg et al., 2006)	Cyclic trace clock and translation invariance	Positive control; no novelty credit and no rooted- h repair
Adelic operator (Knauf, 2013)	Distinct Markov transition operator and spectral problem	Separate phase space
Prescribed-trace counts (Technau, 2023)	Counts of Farey matrix products with fixed trace	No theorem about first-column sums or Liouville phase
Dynamical determinants (Ruelle, 1976; Mayer, 1990, 1991)	Expanding-map, Gauss, and modular transfer-operator formalisms	Comparator only; no proof or owner for Q_s or the rooted return map

The metadata used here is limited to sealed records. In particular, the ASCII author form

“Ozluk” is retained for the two records whose sealed arXiv metadata uses that spelling; no unverified diacritic or issue number is introduced. Ruelle and Mayer are cited solely as changed-model comparators, not as evidence for the present theorem or its novelty.

B.3 Chronology and immutable boundary

The selector was assembled after all six historical cards, their Route outcomes, and the exact witnesses were known. Its unique result is therefore retrospective provenance, not preregistration, outcome-independent evidence, or priority. Paper 39 records candidate existence but does not rank or authorize this successor; Paper 40 neither selects it nor supplies novelty.

The immutable authority release consists of the exact 16-file `preauthority/` package and the two independent-DA files named by the root research pointer. The manuscript, bibliography, figures, writer handoff, diagnostic compilation, and any later canonical integration block remain mutable writer artifacts. No code, result, experiment, evaluation, paper manifest, root README, registry, Git, or mirror artifact is created by this candidate.

Within this boundary the strict conclusion remains `ROUTE_A_REJECTED`, with Route B false. A changed trace clock, enlarged state, non-scalar cocycle, or imported transfer operator requires a new source lock and a fresh Route evaluation; it cannot be used to reinterpret the exact STOP as a positive result for the original object.